

EE6407 Genetic Algorithms and Machine Learning

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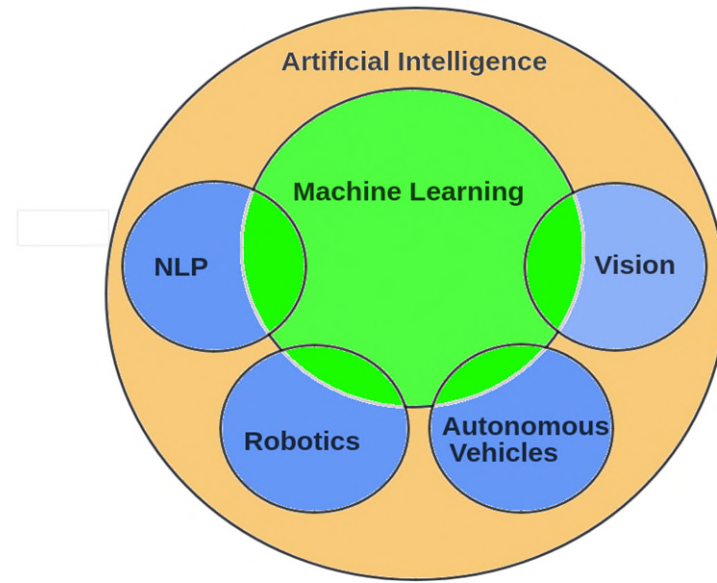
Lecture organization and assessments

- Machine learning lecture
 - ❑ Weeks 5-13
- Continuous Assessments (machine learning)
 - ❑ Total: 30%
 - ❑ Assignments 1&2: 20% (release on Week 7&8, submit by Week 10)
 - ❑ Quiz 2:10%, Tuesday, 11 November 2025, 6:30-7:30pm.
- Exam
 - ❑ Total 60%
 - ❑ 4 questions, 3 from machine learning

AI, Machine Learning, Neural Networks and Deep Learning: at a Glance

- **AI and Machine Learning**

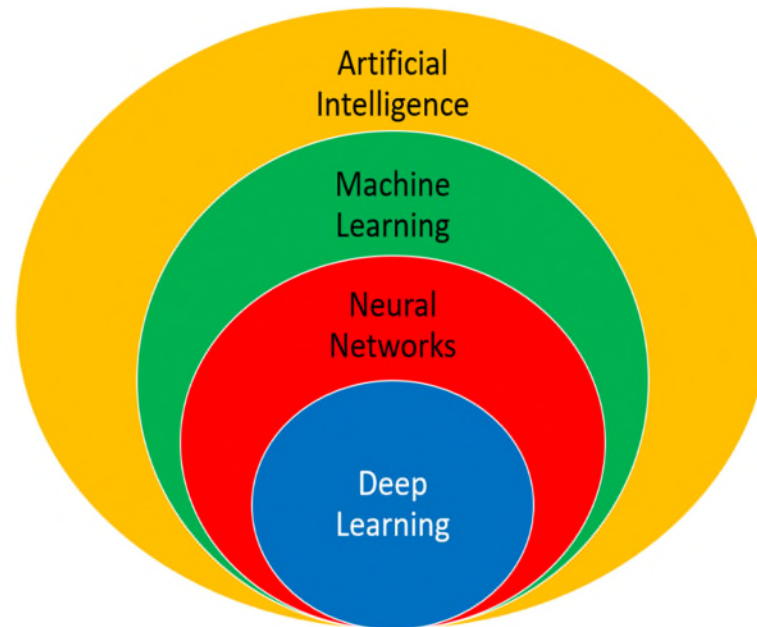
- ❑ AI is a broad term for techniques which enable **machines to mimic human behaviours**
- ❑ Machine learning is a sub-set of AI techniques which enable **machines to learn from experience**
 - Data-driven
 - Statistical algorithms

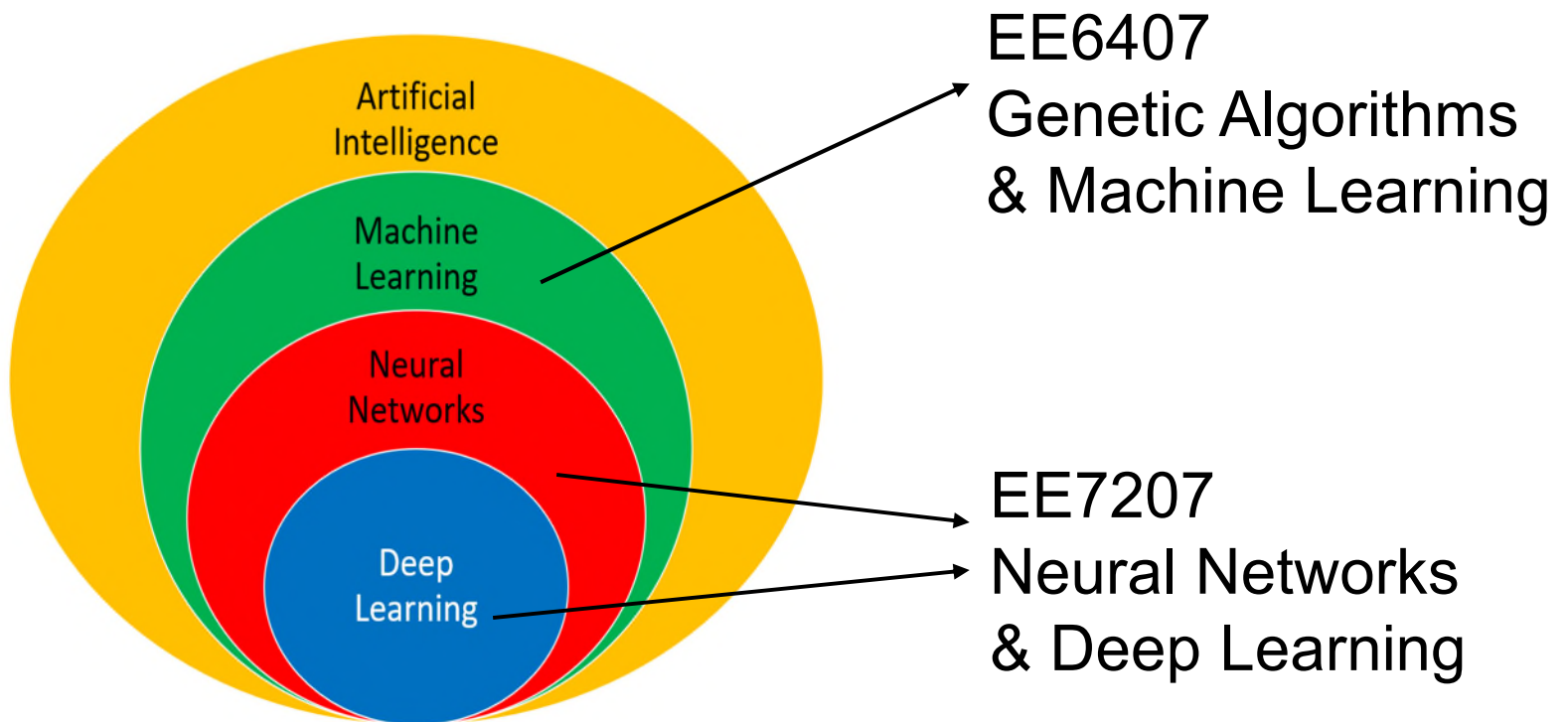


- **Machine Learning, Neural Networks and Deep Learning**

- ☐ Neural networks is a sub-field of machine learning

- ☐ Deep learning is a sub-field of neural networks





Contents

- Introduction to machine learning
- Data preparation for machine learning
- Supervised learning
 - Bayesian decision rule
 - Linear discriminant analysis
 - Support vector machines
 - Decision Trees
 - Dimensionality reduction and feature selection
 - Evaluation of classification
 - Regression
- Unsupervised learning
 - Centroid-based clustering
 - Hierarchical clustering
 - Density-based clustering
 - Distribution-based clustering
 - Evaluation of clustering

References

- R. Duda, P. Harr and D. Stork, Pattern Classification, 2nd edition, Wiley-Interscience.
- Subramanian Chandramouli, Saikat Dutt, Amit Kumar Das., Machine Learning, 1st Edition, 2018, Pearson Education.
- Materials on the Internet.

1. Introduction to Machine Learning

1.1 WHAT IS HUMAN LEARNING?

- ❑ In cognitive science, learning is typically referred to as the process of gaining knowledge through observation.
- ❑ With more learning, tasks can be performed more efficiently.

1.2 TYPES OF HUMAN LEARNING

Human learning happens in one of the three ways:

- ❑ The first type is learning directly under expert guidance.
- ❑ The second type is learning guided by knowledge gained from experts
- ❑ The third type is learning by self or self-learning.

1.2.1 Learning under expert guidance

- ❑ An infant may instil certain traits and characteristics, straight from his/her parents.
- ❑ When starting going to school, the baby is able to learn all these things from his teacher.
- ❑ Then starting higher studies where the person learns about more complex, application-oriented skills. Experts, in general the teachers, who have in-depth subject matter knowledge, help the students in learning these skills.

- ❑ Then the person starts working as a professional in some field. The professional mentors help newcomers in the field to learn on-job.

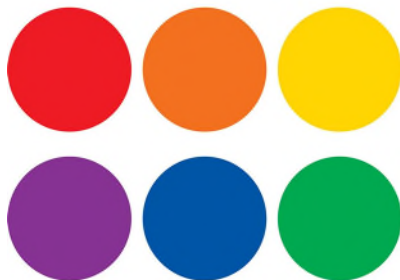
In all phases of life of a human being, there is an element of guided learning, which is the process of gaining knowledge from a person having sufficient knowledge due to the past experience.

1.2.2 Learning guided by knowledge gained from experts

An essential part of learning also happens with the **knowledge learned from experts at some point of time, in some other form/context.**

- A baby can group together all objects of same colour even if his parents have not specifically taught him to do so.

He is able to do so because at some point of time his parents have told him which colour is blue, which is red, which is green, etc



- ❑ A grown-up kid can select one odd word from a set of words because it is a verb and other words being all nouns.

Hospital, Car, Mother, Run

He could do this because of his ability to label the words as verbs or nouns, taught by his English teacher long back.

			
Person	Place	Things	Idea
• David • Emily • He • She • Mother • Father • Grand Father	• Asia • Europe • America • India • Pacific Ocean • Hospital • School	• Mobile • Books • Pencil • Desk • Car • Television • Shoes	• Honesty • Friendship • Freedom • Loyalty • Anger • Wisdom, • Memory

 Crawl	 Run	 Dance	 Dance
 Laugh	 Jump	 Play	 Stand
 Drink	 Write	 Think	 Read
 Sleep	 Clean	 Cry	 Ride
 Kick	 Clap	 Sneeze	 Sing

❑ In a professional role, a person knows to which customers he should market a campaign **from the knowledge about preference that was given by his boss long back.**

In all these situations, there is **no direct learning**. It is **some past knowledge**, shared on some different context, **that is used as a learning to make decisions.**

1.2.3 Learning by self

In many situations, humans are left to learn on their own.

- ❑ A classic example is a baby **learning to walk through obstacles**. He falls down multiple times till he learns that whenever there is an obstacle, he needs to cross over it.
- ❑ He faces the same challenge while **learning to ride a cycle** as a kid or drive a car as an adult.
- ❑ Not all things are taught by others. **A lot of things need to be learnt only from mistakes made in the past.**

1.3 WHAT IS MACHINE LEARNING?

Two fundamental questions:

- Do machines really learn?
- If so, how do they learn?

Tom M. Mitchell, Professor of Machine Learning, School of Computer Science, Carnegie Mellon University:

‘A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .’

This means that a machine can be considered to learn if

- (i) It is able to gather experience by doing a certain task, and
- (ii) It improves its performance in doing the similar tasks in the future.

The **past experience** means **past data related to the task**.

- ❑ In learning to play checkers, experience represents the **experience of playing checkers**
- ❑ In image classification, experience represents the **past data with images having labels**

1.3.1 How do machines learn?

The basic machine learning process can be divided into three parts:



- ❑ **Input Data:** Past data or information is utilized as a basis for future decision-making
- ❑ **Abstraction:** The input data is represented in a broader way through the underlying algorithm
- ❑ **Generalization:** The abstracted representation is generalized to form a framework for making decisions

1.3.1.1 Abstraction

Abstraction aim to derive a conceptual map based on the input data. This concept map is called a model in the machine learning paradigm.

The model may be in any one of the following forms:

- ❑ Computational blocks like if/else rules
- ❑ Mathematical equations
- ❑ Specific structures like trees or graphs
- ❑ Logical groupings of similar observations

The choice of the model for a specific learning problem is a human task, based on multiple aspects such as:

- ❑ **The type of problem to be solved:** Whether the problem is related to forecast or prediction,
- ❑ **Nature of the input data:** How exhaustive the input data is, the data types, etc.
- ❑ **Domain of the problem:** If the problem is in a critical domain, with a high rate of data input, and need for immediate inference

Once the model is chosen, the next task is to **fit the model based on the input data**. Assume a simple model is represented by the following equation:

$$y = c_1x + c_2$$

Based on the input data, we will find out the values of c_1 and c_2 . So, **fitting the model**, in this case, **means finding the values of the unknown coefficients of the model**.

This **process of fitting the model** based on the input data is known as **training or learning**.

The input data is known as **training data**.

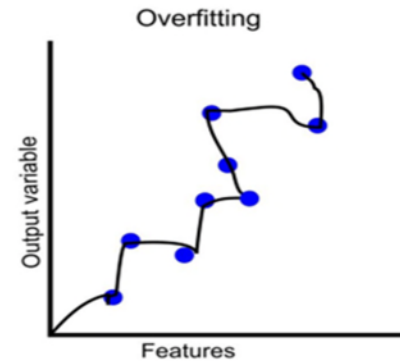
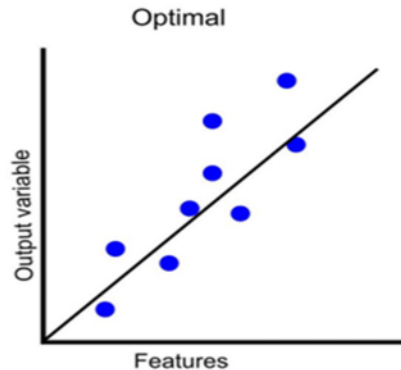
1.3.1.2 Generalization

Generalization is to tune up the abstracted knowledge to a form which can be used to make future decisions.

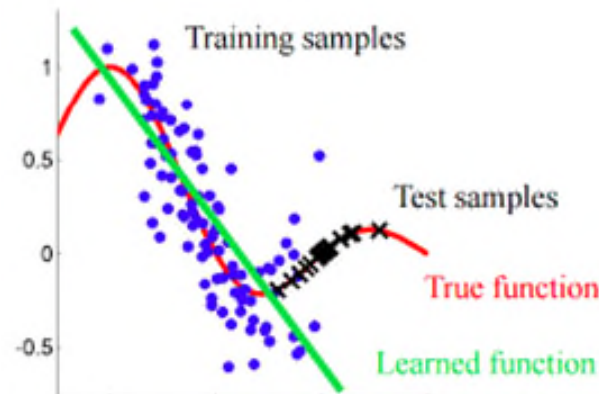
The model is trained based on a finite set of data, which may possess a limited set of characteristics.

When we apply the model to make decision on a set of unknown data, usually termed as test data, we may encounter two problems:

- ❑ The trained model is **aligned with the training data too much**, hence may not portray the actual trend.



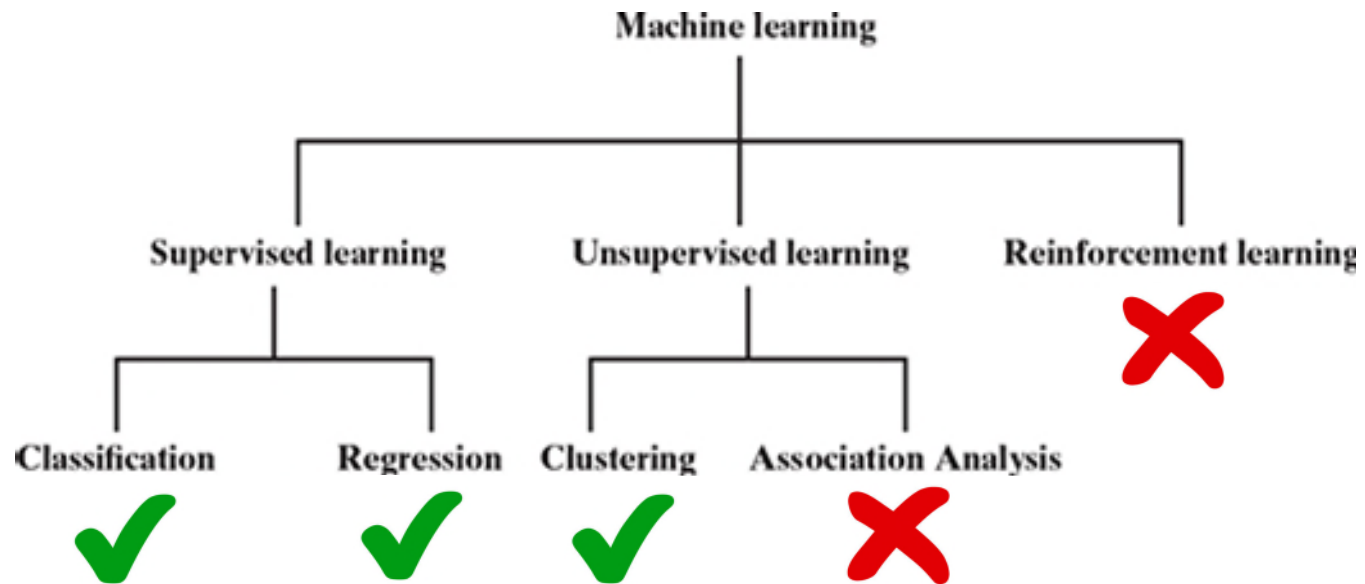
- ❑ The test data possess certain **characteristics apparently unknown** to the training data.



1.4 TYPES OF MACHINE LEARNING

Machine learning can be classified into three broad categories:

- ❑ **Supervised learning** – Also called predictive learning. A machine **predicts the class** of unknown objects based on prior class-related information of similar objects.
- ❑ **Unsupervised learning** – Also called descriptive learning. A machine **finds patterns** in unknown objects by grouping similar objects together.
- ❑ **Reinforcement learning** – A machine **learns to act on its own** to achieve the given goals.



1.4.1 Supervised learning

The major motivation of supervised learning is **to learn from past information**.

In context of machine learning, this **past information is the experience**.

For example, the task is to segregate the images by either shape or colour of the object.

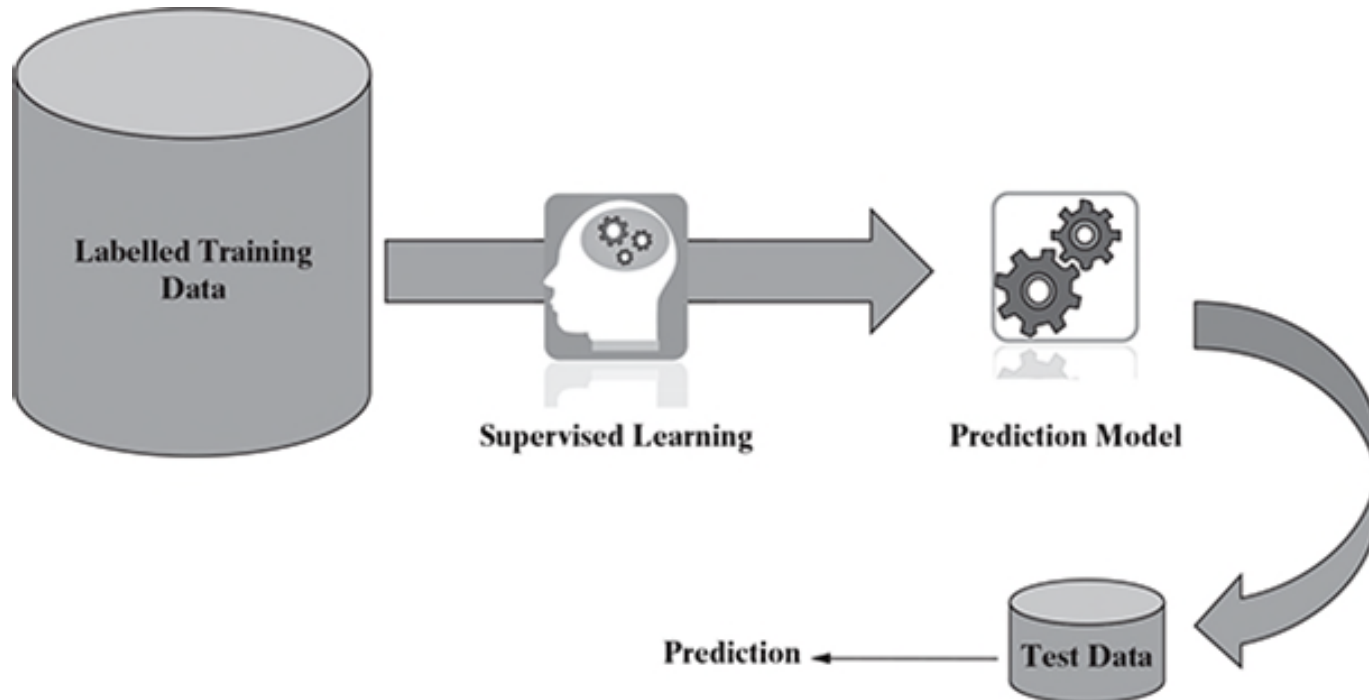
- ❑ If **by shape**, the images of round-shaped objects need to be separated from images of triangular-shaped objects.
- ❑ If **by colour**, images of blue objects need to be separated from images of green objects.

But how can the machine know

- ❑ What is round shape, or triangular shape?
- ❑ What is blue or green in colour?

A machine needs to be provided with the basic information on shape and colour.

- ❑ This basic input is given in the form of training data.
Training data is the past information on a specific task.
- ❑ Training data will have past data, along with a tag on whether the image is round or triangular, or blue or green in colour.
- ❑ The tag is called label, and we say that the training data is labelled in case of supervised learning.



Supervised learning

Some examples of supervised learning are:

- ❑ Predicting the results of a game
- ❑ Predicting whether a tumour is malignant or benign
- ❑ Predicting the price of real estate, stocks, etc.
- ❑ Classifying texts such as spam or non-spam

Let's consider two of the above examples: 'predicting **whether a tumour is malignant or benign**', and 'predicting **price of real estate**'.

Are these two problems the same in nature?

The answer is 'no'.

- ❑ Case 1: predict **which category or class** an unknown data belongs to.
- ❑ Case 2: **predict a real value** and not a class.

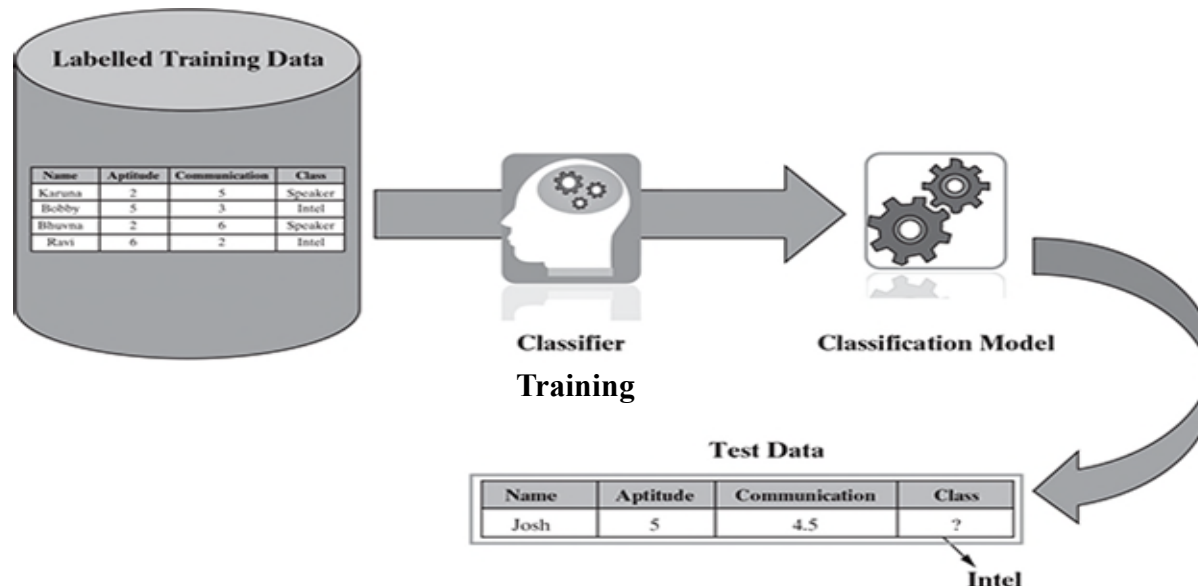
1.4.1.1 Classification

If the target objective of predicting is **to assign a class label**, this type of problem is referred as **classification problem**.

Some typical classification problems include:

- ☐ Image classification
- ☐ Prediction of disease
- ☐ Win-loss prediction of games
- ☐ Prediction of natural calamity like earthquake, flood, etc.
- ☐ Recognition of handwriting
- ☐ Face recognition
- ☐ etc.

The following figure depicts the typical process of classification.

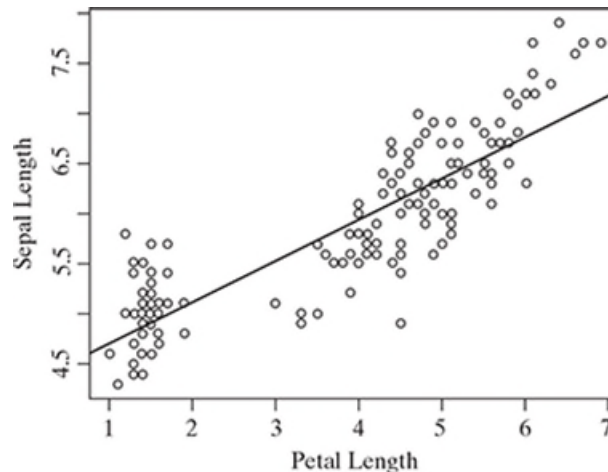


In summary, classification is a type of supervised learning where a target variable, which is categorical, is predicted for test data. The target categorical variable is known as class.

1.4.1.2. Regression

In regression, the objective is to predict numerical (real-valued) variables.

In case of linear regression, a straight-line relationship is fitted between the predictor variables and the target variables



Typical applications of regression can be seen in

- (1) Demand forecasting in retails
- (2) Sales prediction for managers
- (3) Price prediction in real estate
- (4) Growth prediction in economics
- (5) Enrol student number prediction in education

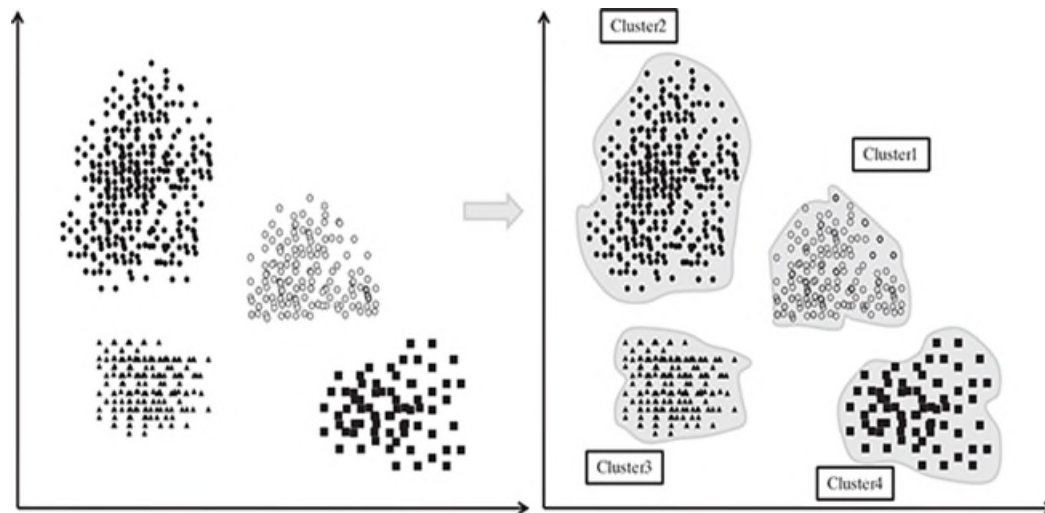
1.4.2 Unsupervised learning

In unsupervised learning, there is **no labelled training data to learn from** and **no prediction to be made**.

- ❑ The objective of unsupervised learning is to find **natural groupings or patterns** within the data elements or records.
- ❑ Unsupervised learning is often termed as **descriptive learning**
- ❑ The process of unsupervised learning is referred as **pattern discovery or knowledge discovery**

Clustering is the main type of unsupervised learning. It intends **to group or organize similar objects together**:

- ❑ Objects belonging to the same cluster are quite similar to each other;
- ❑ Objects belonging to different clusters are quite dissimilar.



One variant of unsupervised learning is **association analysis**, where the **association between data elements is identified**.

Applications of association analysis include market basket analysis and recommendation etc.

TransID	Items Bought
1	{Butter, Bread}
2	{Diaper, Bread, Milk, Beer}
3	{Milk, Chicken, Beer, Diaper}
4	{Bread, Diaper, Chicken, Beer}
5	{Diaper, Beer, Cookies, Ice cream}
...	...

Market Basket transactions

Frequent itemsets → (Diaper, Beer)

Possible association: Diaper → Beer

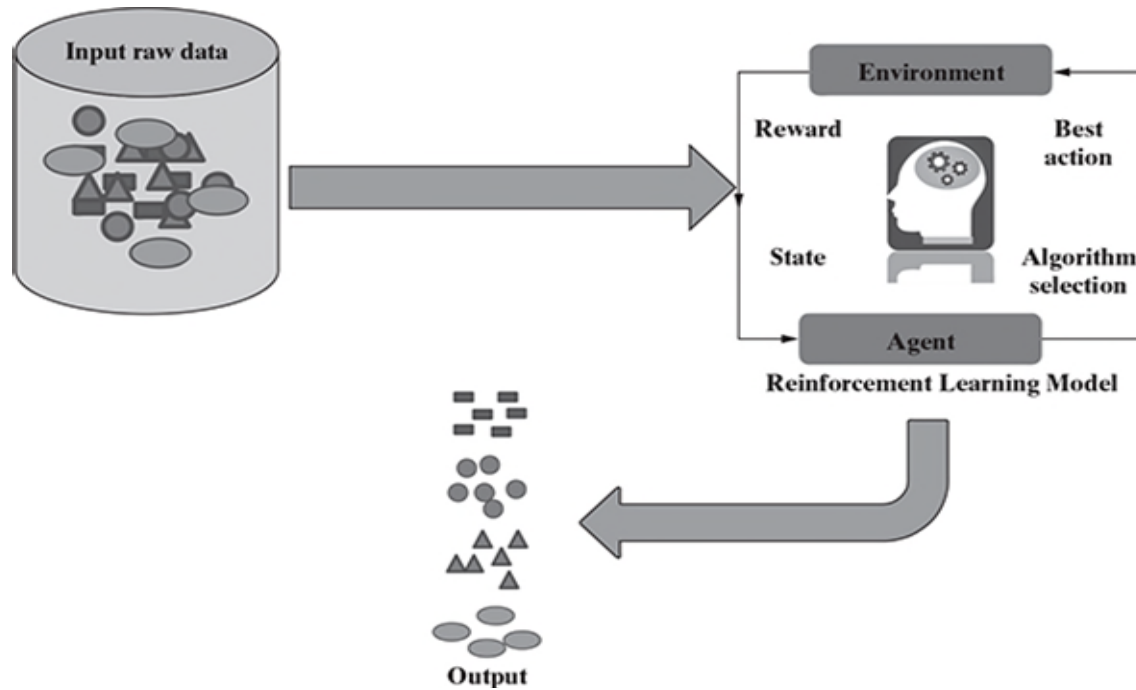
1.4.3 Reinforcement learning

Let's try to understand reinforcement learning through the example of the child learning to walk.

- ❑ When the child is able to walk smoothly, a reward (such as claps) is given.
- ❑ When the child falls down, their parents do not give claps (no reward).
- ❑ Slowly a time comes when the babies learn from mistakes and are able to walk with much ease.

This process of learning is known as reinforcement learning. An example is self-driving cars.

The following figure shows the high-level process of reinforcement learning.



1.5 APPLICATIONS OF MACHINE LEARNING

- **Banking and finance**

- ☐ Fraudulent transactions detection
- ☐ Customer churn prediction

- **Insurance**

- ☐ Risk prediction
- ☐ Claims management

- **Healthcare**

- ☐ Health condition monitoring/prediction
- ☐ Disease diagnosis

1.6 STATE-OF-THE-ART LANGUAGES/TOOLS IN MACHINE LEARNING

- Python

- ❑ Python is one of the most popular, open-source programming language widely adopted by machine learning community.
- ❑ Python has very strong machine learning library named **scikit-learn**, which has various classification, regression, and clustering algorithms embedded in it.

- R

- ❑ R is a language for statistical computing and data analysis. It is an open-source language, extremely popular in the academic and industry.
- ❑ R is a very simple programming language with a huge set of libraries available for machine learning.

- MATLAB

- ❑ Being proprietary software, MATLAB is developed much more professionally, tested rigorously, and has comprehensive documentation.
- ❑ MATLAB also provides extensive support of statistical functions and has a huge number of machine learning algorithms in-built.

1.7 ISSUES IN MACHINE LEARNING

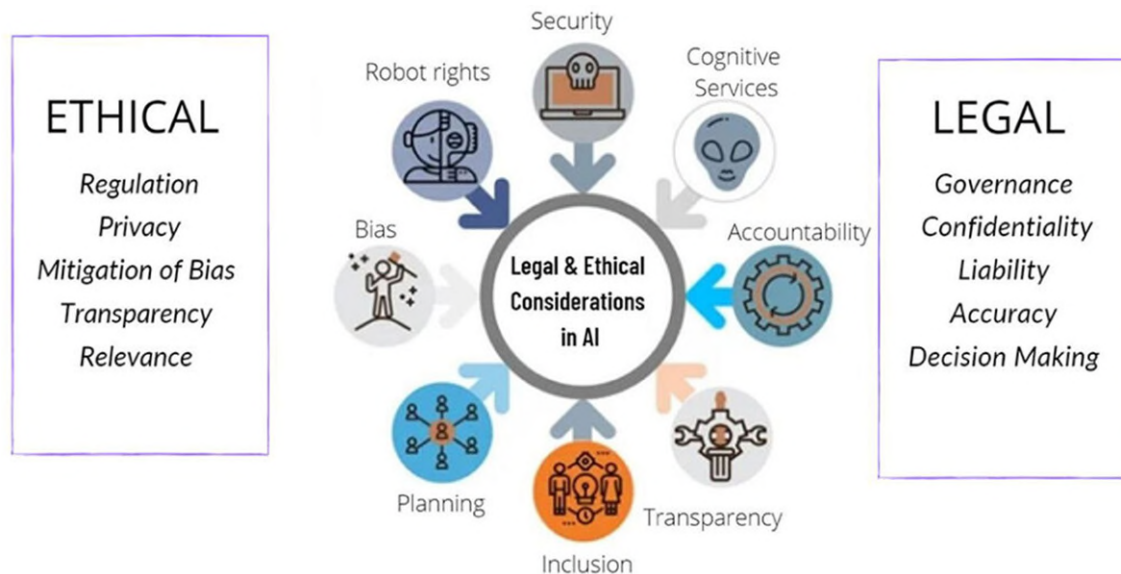
The laws and regulations, cultural background, emotional maturity of people differ drastically in different countries.

All these factors make the use of machine learning and the issues originating out of machine learning usage are quite different.

- ❑ The biggest fear and issue arising out of machine learning is related to privacy and the breach of it.



- ❑ Even if there is no breach of privacy, there may be situations where actions were taken based on machine learning may create an **adverse reaction**.
- ❑ A very critical consideration before applying machine learning is that **proper human judgement** should be exercised.



1.8 SUMMARY

Machine learning imbibes the philosophy of human learning, i.e. **learning from expert guidance** and **from experience**.

(1) The basic machine learning process can be divided into three parts.

- ❑ **Data Input:** Past data or information is utilized as a basis for future decision-making.
- ❑ **Abstraction:** The input data is represented in a summarized way.
- ❑ **Generalization:** The abstracted representation is generalized to form a framework for making decisions.

(2) Machine learning can be classified into three broad categories:

- ❑ **Supervised learning**: The objective of this learning is to predict class/value of unknown objects based on prior information of similar objects.
- ❑ **Unsupervised learning**: This learning aims to find groups or patterns in unknown objects by grouping similar objects together.
- ❑ **Reinforcement learning**: A machine learns to act on its own to achieve the given goals. Examples: self-driving cars, intelligent robots, etc.

(3) Machine learning has been adopted by various industry domains.

(4) Some of the most adopted platforms to implement machine learning include Python, R, MATLAB etc.

(5) To avoid ethical issues, the critical consideration is required before applying machine learning and using any outcome from machine learning.